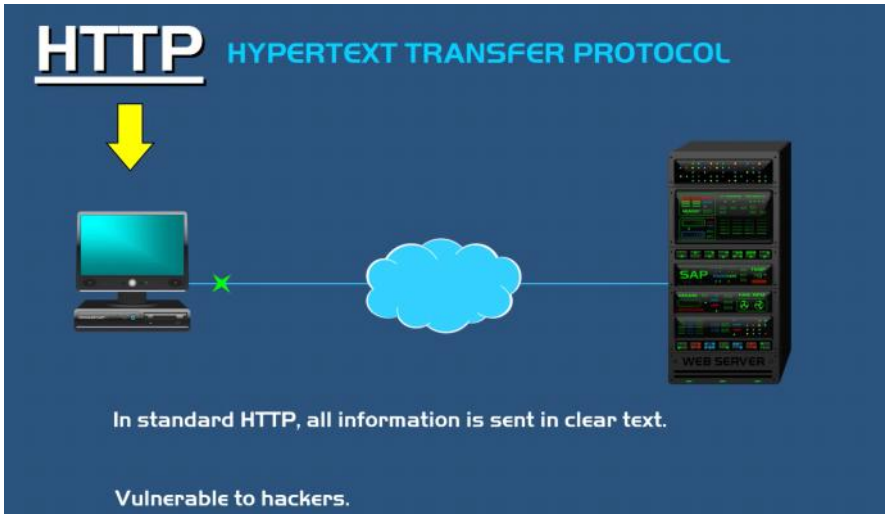
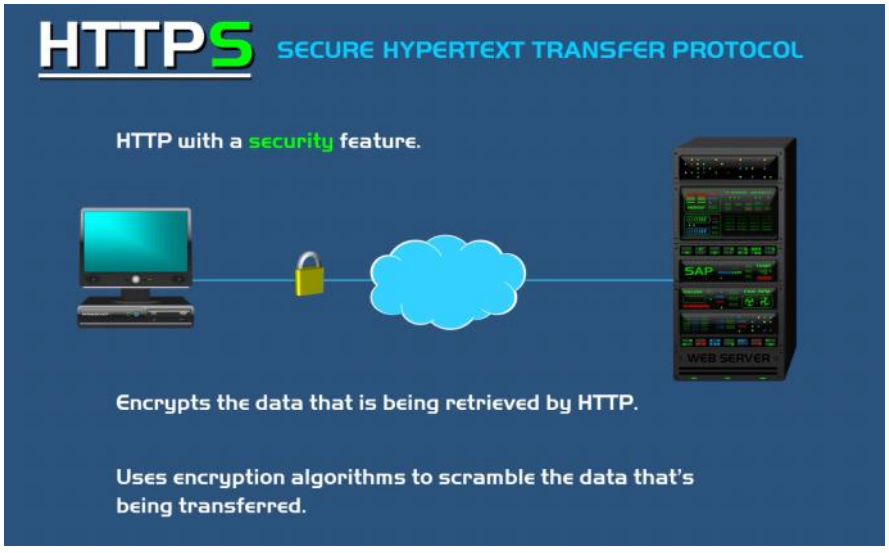
SSL, TLS, HTTP, HTTPS

Sunday, April 5, 2026 5:03 PM

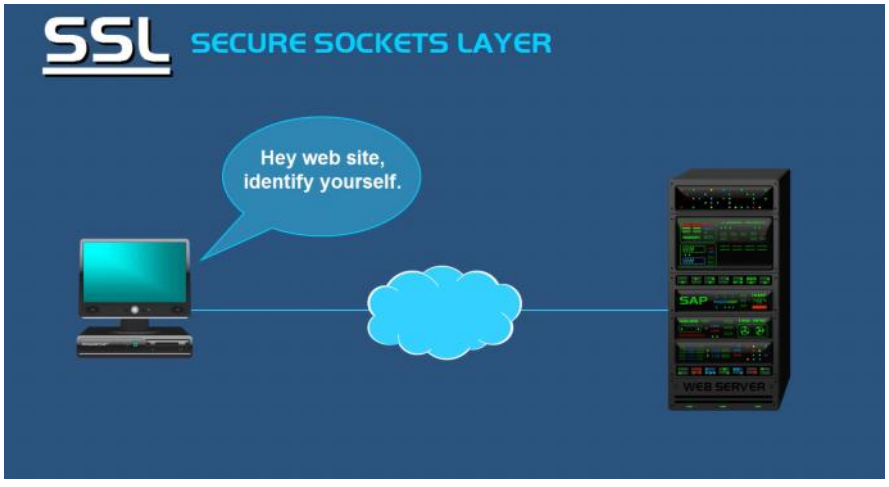
HTTP (Hypertext Transfer Protocol) is the standard protocol used to view web pages. Data sent via HTTP is in clear text, making it vulnerable to hackers who can intercept sensitive information like passwords or credit card details.

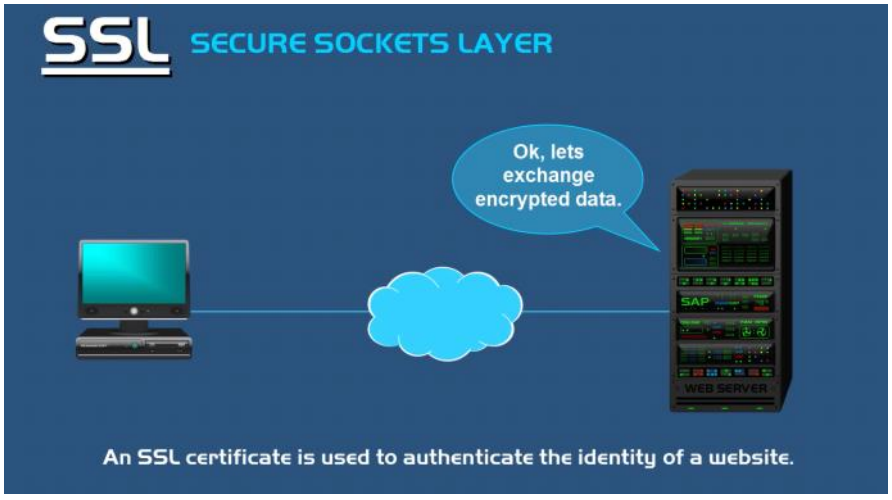


HTTPS (Secure Hypertext Transfer Protocol) is the secure version of HTTP. It uses encryption algorithms to scramble data, rendering it unreadable to anyone who might try to steal it. When a website is secure, browsers typically display a padlock symbol in the address bar.



HTTPS relies on two main security protocols:





SSL (Secure Sockets Layer): Uses public key encryption to authenticate a website's identity via an SSL certificate. This ensures the connection between your computer and the server is trustworthy.

TLS (Transport Layer Security): The modern, industry-standard successor to SSL, built on similar principles to provide authentication and data encryption.
Today, most major websites use HTTPS by default to protect user data and maintain favorable Google search rankings, as Google flags unencrypted sites as insecure.